

Theorem 6(b): $x(x+y) = x$

Proof: $x(x+y) = x \cdot x + x \cdot y$ (Postulate 4(a))
 $= x + x \cdot y$ (Theorem 1(b))
 $= x$ (Theorem 6(a)) $\square$

Note: The following table must be remembered for further applications.

Table 2-1 Postulates and theorems of Boolean algebra

Postulate 2	(a) $x+0 = x$	(b) $x \cdot 1 = x$
Postulate 5	(a) $x+x' = 1$	(b) $x \cdot x' = 0$
Theorem 1	(a) $x+x = x$	(b) $x \cdot x = x$
Theorem 2	(a) $x+1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3, involution	$(x')' = x$	
Postulate 3, commutative	(a) $x+y = y+x$	(b) $x \cdot y = y \cdot x$
Theorem 4, associative	(a) $x+(y+z) = (x+y)+z$	(b) $x(yz) = (xy)z$
Postulate 4, distributive	(a) $x(y+z) = xy + xz$	(b) $x+yz = (x+y)(x+z)$
Theorem 5, De Morgan	(a) $(xy)' = x'y'$	(b) $(x+y)' = x'y$
Theorem 6, absorption	(a) $x+xy = x$	(b) $x(x+y) = x$

• We shall give validity proofs of Theorems 4 and 5 using Truth Tables since their algebraic proofs are very long.

Validity of Theorem 4(a):

x	y	z	$(y+z)$	$x+(y+z)$	$x+y$	$(x+y)+z$
0	0	0	0	0	0	0
0	0	1	1	0	0	1
0	1	0	1	0	1	1
0	1	1	1	0	1	1
1	0	0	0	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1